

Workshop: Differentiation Chain Rule

UEA LEARNING ENHANCEMENT TEAM MATHEMATICS & STATISTICS

$$y = [f(x)]^n$$

These are expressions that look like: $y = (5x^2 + 3x + 1)^4$

The function $5x^2 + 3x + 1$ is inside a 'to the power 4' function.

Define a 'dummy' variable (say u) to represent the 'inside' function:

$$u = f(x), \text{ so now } y = u^n$$

The chain rule can be applied:

$$\frac{dy}{dx} = \frac{dy}{du} \times \frac{du}{dx}$$

So:

$$\frac{dy}{du} = nu^{n-1}, \quad \frac{du}{dx} = f'(x) \quad (f'(x) \text{ is the derivative of the function } f(x))$$

$$\frac{dy}{dx} = nf'(x)[f(x)]^{n-1}$$

For the example above:

$$\text{Let } u = 5x^2 + 3x + 1, \text{ so } y = u^4$$

$$\frac{dy}{du} = 4u^3, \quad \frac{du}{dx} = 10x + 3$$

$$\frac{dy}{dx} = 4u^3(10x + 3) = 4(10x + 3)(5x^2 + 3x + 1)^3$$

Find the derivative of:

1. $y = (5x + 4)^5$

2. $y = (5x^3 - x - 1)^9$

3. $y = (4x + 9)^{-1}$

$$4. y = \sqrt{(8x + 1)}$$

$$5. y = (e^x - 2x)^4$$

$$y = e^{f(x)}$$

These are expressions that look like $y = e^{x^2+4x}$. Here the $x^2 + 4x$ is inside the exponential function.

Use the dummy variable $u = f(x)$, so that $y = e^u$.

The chain rule can be applied:

$$\frac{dy}{dx} = \frac{dy}{du} \times \frac{du}{dx}$$

So:

$$\frac{dy}{du} = e^u = e^{f(x)}, \quad \frac{du}{dx} = f'(x)$$

$$\frac{dy}{dx} = f'(x)e^{f(x)}$$

For the example above:

$$\begin{aligned} u &= x^2 + 4x, \quad y = e^u \\ \frac{dy}{du} &= e^u = e^{x^2+4x} \quad \frac{du}{dx} = 2x + 4 \\ \frac{dy}{dx} &= (2x + 4)e^{x^2+4x} \end{aligned}$$

Find the derivative of:

$$6. y = e^{4x}$$

$$7. y = 5e^{x^2}$$

$$8. y = 2e^{x^3+2x-4}$$

$$9. y = e^{4-x-x^2}$$

$$10. y = 6e^{1+\frac{1}{x}}$$

Mixed

Use the examples above and your understanding of the chain rule to differentiate:

$$11. y = (8x + 4)^{-4}, \quad \frac{dy}{dx} =$$

$$12. y = e^{1+\ln x} \quad \frac{dy}{dx} =$$

$$13. y = \ln(x^2 + 3), \quad \frac{dy}{dx} =$$

$$14. y = 5t + e^{6t-1}, \quad \frac{dy}{dt} =$$

$$15. S = 4 \ln(5x^2 + 2x^4) + (x - 1)^5, \quad \frac{dS}{dx} =$$

$$16. P = e^{(3t+5)^6}, \quad \frac{dP}{dt} =$$